Tracking and Analyzing Use of Evidence in Public Policymaking Processes:

A Theory-Grounded Content Analysis Methodology

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Key Messages:

- Studies about policymakers' use of research evidence frequently track 'evidence' as opposed to tracking 'use'.
- Policymakers use evidence to inform their thoughts and decisions but also to influence those of others.
- Policymakers' communication with and about evidence is a reliable proxy of how and for what purpose they use evidence.
- Methods from the field of persuasive communication can yield a rich and nuanced account of policymakers' evidence use.

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Background

Evidence-informed decision-making has long been considered the gold standard of sound policy and practice. In theory, an evidence-informed approach to policymaking enables policymakers to base their decisions and actions on good information, to ask critical questions when they are presented with evidence for or against a particular policy, and to evaluate the success or failure of policies they implement (Oxman et al., 2009). In reality, however, use of research evidence in public policymaking appears to be infrequent, inconsistent, and at times misinformed (National Research Council, 2012, Nutley et al., 2007, Oliver et al., 2014). Past research on the causes of this research utilization gap has emphasized the importance of improving communication and building relationships between scientists and policymakers, including making research more accessible to policymakers by providing clear, concise, and timely research summaries that are tailored to their specific needs and preferences (van de Goor et al., 2017, Oliver et al., 2014). Theory and research on this topic has since evolved to recognize that use of research evidence in policymaking bears little resemblance to the orderly, systematic and calculated process often portrayed in the literature on the topic. Rather, use of research evidence unfolds within a complex web of relationships and interactions (Tseng, 2012, Moat et al., 2013, Boaz, 2013), with research and non-research evidence entering the policymaking process through intermediaries such as news media, personal acquaintances, lobbyists, and constituents (Oliver and de Vocht, 2017), and evidence being frequently utilized for enlightenment or political purposes rather than for informing policy decisions directly (Boswell, 2008, Kingdon, 2011, Pearce et al., 2014).

This contemporary view of policymakers' use of evidence, we argue, presents a significant methodological challenge to scholars interested in tracking and analyzing the use and impact of research evidence in public policymaking processes. Past studies on this topic (e.g., Castellani et al., 2016, Gormley, 2011, Meagher et al., 2008, Moat et al., 2013, Ritter and Lancaster, 2013, Zardo and Collie, 2014) have commonly employed methodologies that are designed to track research evidence as it moves from research producers to users, for example, by interviewing policymakers about their acquisition, interpretation, and use of research evidence (Bogenschneider and Corbett, 2010) or by tracking citations to research in policy documents (Benoit and Herzog, 2017, Goldie et al., 2014). In this regard, the application of more complex methods such as social network analysis has expanded the ability to track the diffusion of research evidence within larger systems such as school districts and advocacy coalitions (Finnigan et al., 2013, Neal et al., 2015), although limitations remain. For one, direct or explicit references to sources of research evidence are rarely included in policy documents and cannot be readily recalled from memory by policymakers when they are asked to do so (Haunschild and Bornmann, 2017). For another, tools such as bibliometric analysis are not well-equipped to detect and track the use of non-research evidence by policymakers (Epstein et al., 2014, Pearce et al., 2014), nor are such methods appropriate to determine how and why evidence is used in a particular context (Weiss, 1979). Still, in our view, the most significant limitation of such methodologies as they are currently implemented is that they focus on tracking and analyzing *evidence* as opposed to tracking and analyzing *use*. In that sense, existing methods potentially detract attention away from the need to better conceptualize and operationalize ‘use’ in the specific context of public policymaking.

Weiss (1979) attempted to address this issue early on when she pioneered the familiar distinction among instrumental use (i.e., direct and specific application of research evidence in decisionmaking processes), conceptual use (i.e., use of research for general enlightenment), and symbolic use (i.e., use of research to legitimatize and sustain predetermined positions). Davies, Nutley, and Walter (2008) proposed, in addition, that research evidence use is context-dependent and that, therefore, 'use' ought to be defined and studied relative to local priorities, culture, and organizational structures. Despite these valuable insights, use of research evidence in public policymaking remains largely under-conceptualized and under-operationalized in the current literature. We believe this is due to 'use' being too narrowly understood and operationalized as the cognitive process by which research evidence informs the thoughts, decisions, and actions of individual policy decisionmakers (which is what available instruments are designed to measure, e.g., Makkar et al., 2016, Palinkas et al., 2016), whereas policymakers routinely use evidence, strategically, to influence the thoughts, decisions, and actions of others (Boswell, 2008, Majone, 1989). Since evidence use is necessarily embedded in common practices (Freeman et al., 2011), and given that policymaking-related practices primarily involve persuasion and bargaining with other actors (Kingdon, 2011, Majone, 1989), it is important to recognize that ideas, knowledge, and evidence play a role in the policymaking process to the extent that policymakers' consider them persuasive. Boswell (2008) makes this point more explicitly by arguing that expert knowledge serves two important symbolic functions in public policymaking: enhancing the legitimacy of an organization and its claim to resources or jurisdiction over particular policy areas (a legitimizing function), and lending authority to particular policy positions (a substantiating function), particularly in cases of political contestation.

A Persuasion Perspective on Use of Evidence in Public Policymaking

Accepting the premise that use of research evidence in public policymaking is deliberate, strategic, and inherently political, we propose that theories of persuasion and argumentation provide a solid basis for conceptualizing 'use' because they explain how the way evidence is presented is likely to influence the way audiences comprehend, interpret, evaluate, and ultimately accept or reject a certain claim or position (O'Keefe, 2002, Petty and Cacioppo, 1981). Message framing theory (see Scheufele and Tewksbury, 2007), in particular, has been extensively used to this end, although two conceptually distinct variants of framing are often confused in the literature (Cacciatore et al., 2016). One variant, originally proposed by prospect theory (Tversky and Kahneman, 1981), considers how contextualizing information or facts in a particular way, for example in terms of potential gain versus potential loss, can influence the choices people make. A second variant views framing primarily as a mechanism for promoting a specific interpretation of or attribution about a certain object (e.g., a person, an issue, or a proposed policy) by emphasizing some information or facts about an issue (Goffman, 1974), for example through the use of metaphors (Schlesinger and Lau, 2000). Both variants draw on the notion of mental schema, the pre-existing knowledge, beliefs, and values that people use to evaluate new information and to generate attributions or judgments about objects or issues.

We propose framing as an intuitive way to conceptualize and operationalize use of evidence in public policymaking, including clarifying the mechanisms underlying the typology of use (instrumental, conceptual, and symbolic) popularized by Weiss (1979) that scholars have generally struggled to operationalize (Boswell and Smith, 2017). For example, enlightenment (or learning) is often referenced as the mechanism that underlies conceptual use of research evidence by policymakers, despite little evidence that supports this notion (Boswell and Smith, 2017). Framing theory would suggest instead that conceptual use of research evidence likely involves policymakers' reliance on evidence to contextualize information or a claim in a way that fits with an existing mental schema, or an organizing structure. Such a mental scheme or structure is then routinely applied to make sense of issues, for example, organizing information according to causes, consequences, or solutions to public problems (Kingdon, 2011). Therefore, it is logical to assume that evidence that is relevant to these dimensions will be evaluated against what policymakers already know and believe about the problem. Similarly, a framing perspective would understand symbolic (or tactical) use of research evidence as a deliberate attempt to influence what others consider or think regarding an issue, either by selectively emphasizing evidence that substantiates or reinforces a certain position or conclusion or suggesting that the evidence provided is either authoritative, ambiguous, or false (Majone, 1989). Instrumental use of research evidence in this perspective would involve use of evidence to compare two or more alternative ways of understanding problems and solutions and to demonstrate that the available research evidence is logically consistent with one of these alternatives.

Thus, a persuasion approach to use of research evidence recognizes that research and non-research evidence are influential in public policymaking processes only to the extent they are connected to arguments and that evidence may be used for different purposes and for different motivations (see also Boswell, 2008, Majone, 1989). It follows that we can reliably infer motivations for using research evidence by analyzing the structure of arguments. According to argumentation theory (Toulmin, 2003, Jackson and Jacobs, 1980), persuasive arguments are composed of three basic elements: claim, evidence, and a warrant (an explanation of why or how the evidence supports the claim). For example, "recent tuition increases at most state universities make it increasingly difficult for people from lower-income families to obtain a college degree (evidence), thus we should cap tuition increases (claim) in order to maintain our national commitment to make public education accessible to students from all economic backgrounds (warrant or logical link)" (Stiff and Mongeau, 2003, p. 129). Arguments that are structured such that evidence is presented first followed by a claim and a warrant, as in the example above, are indicative of instrumental use of research evidence because such arguments state a conclusion that is logically derived from evidence. In contrast, when an argument is structured such that a specific conclusion or claim appears first and evidence that presumably substantiates this claim follows, we can infer symbolic or tactical use of research evidence since evidence is used to prescribe, rather than suggest, a particular conclusion. Lastly, conceptual use of research evidence can be inferred when evidence is presented to support an argument about some aspect of objective reality (e.g., causes or consequences of the problem) without drawing an explicit conclusion or interpretation, but rather inviting audiences to draw an obvious conclusion or interpretation, which may reinforce or contradict their existing knowledge, understanding, and beliefs regarding the issue.

In this article, our primary objectives are to describe a content analysis methodology we developed based on the persuasion and argumentation framework we outlined above, and to demonstrate how it can support a more complete and nuanced analysis of policymakers' use of evidence. We illustrate the application of this methodology to the analysis of evidence-supported arguments in policy documents, which we define as official documents that contain a detailed description of procedures, deliberations, decisions, or actions relevant to proposing, implementing, and/or evaluating a particular public policy. One of the important advantages of working with policy documents is that there are few aspects of the policymaking process that do not involve the verbal or written use of language that is recorded and archived (Bowen, 2009, Caulley, 1983). In addition to reflecting the thoughts and actions of individual policymakers (Benoit and Herzog, 2017, Bowen, 2009, Freeman and Maybin, 2011), policy documents capture the interactions policymakers engage in as they debate public policies, including how they present and interpret evidence (Altheide, 2000). By analyzing policymakers' use of evidence in this context, we gain a more complete understanding of what, when, and how evidence is used in public policymaking processes. The example we provide is taken from a research initiative that tracked and analyzed policymakers' evidence use in the formulation of federal policies to combat childhood obesity in the United States from 2000 to 2014.

Methods

In general, content analysis methodology involves a sequential process of selecting, retrieving, and apprising relevant documents; developing and testing a coding instrument; training coders and assessing intercoder reliability; applying the coding scheme to extract quantitative and qualitative data; and analyzing the data using standard statistical methods (Krippendorff, 2013). While we outline and explain the process we employed in this specific research project, we recognize that elements such as the selection of relevant documents and data analyses are determined by the specific goals and circumstances of each study. For this reason, our primary focus is on describing the content analysis instrument we developed to systematically measure use of research and non-research evidence.

Content Analysis Instrument

The content analysis instrument we developed (see Appendix) combines elements of content and thematic analysis to track use of research and non-research evidence in arguments (the basic unit of analysis). The instrument was developed and validated through several iterations. The initial version was developed based on a scoping review of the literature across multiple fields (political science, public health, communication, persuasion, and use of research evidence) for existing content analysis instruments. In the next step, the original list of variables we compiled through the literature review was modified (with some items added and other removed) based on inputs from experts and key-informant interviews with individuals involved in policymaking (mostly legislative staff of elected officials but also government bureaucrats). We next applied the modified coding scheme to a random selection of policy documents to identify additional potentially useful variables that we can include as well as variables that could not be systematically coded based on the information included (e.g., the quality of the research referenced or the exact source of research). Finally, we had naïve coders use the instrument to ensure its application is intuitive and systematic across coders. We refined the final version of the instrument according to the results of these additional tests and after establishing intercoder reliability (see description below).

The instrument is composed of two major sections. First, the content analysis component is designed to extract and organize information regarding the *who* (supplier of evidence), *what* (type and source of evidence used), *where* (the specific policymaking context in which evidence is used), and *when* (timing) of evidence use. This is because factors such as source credibility and competence, the perceived relevance, accuracy, and legitimacy of the information presented, and the context and timing of delivering information were consistently found to be important in increasing or decreasing the likelihood that information is processed and used (O'Keefe, 2002). Whereas such factors are more directly measured by interviewing members of the audience (policymakers, in this case), they can also be reliably inferred from observing patterns in documents, such as the frequency in which certain types of evidence or suppliers of evidence appear (Krippendorff, 2013). In fact, exploring such patterns independent of users' perspective may generate additional insights about routine ways or circumstances under which evidence is engaged that users are not fully conscious of or can easily recall from memory (Finnigan et al., 2013, Goldie et al., 2014, Feldman, 2013).

Second, the thematic analysis component of our content analysis instrument (see Appendix for list and details of specific variables) is designed to extract information regarding the use of evidence in policy arguments based on the theoretical framework we outlined above. Specifically, it includes coding for the *goal of using evidence* (e.g., highlight the objective status of a problem, speak to causes of the problem, etc.), the *valance* (i.e., whether evidence is principally used to support or contradict a specific claim or conclusion), and the *motivation* for using research evidence (instrumental, conceptual, or tactical use, further broken down by what specific claim is being established, e.g., suggesting that a certain solution is feasible or not). Clearly, applying these codes correctly requires a careful consideration of the specific context in which arguments are made, which is why we opted to use manual coding. Specially, coders were trained to first read the entire document so they have a better sense of the dynamics of the interactions taking place and the focus of interactions before coding each argument. When there was ambiguity about the appropriate codes to apply, coders consulted senior project staff and agreement was reached through review and discussion.

One final set of codes was designed to capture the policy context in which evidence is used, since it is likely that some form or type of evidence will be more influential in some policy context (e.g., policy agenda-setting) but less influential in others (e.g., policy formulation). The meta-data associated with each document such as the date of publication, type of policy document (e.g., bill, hearings, floor debate, report, speech, statement, or a proclamation), the policy system (e.g., House or Senate) or subsystem (e.g., congressional subcommittee) associated with the document, and the specific policymaking focus captured in the document (e.g., legislation, oversight, investigation, authorization, etc.) are particularly useful to establish the context of evidence use. In this way, it is also possible to reconstruct the actual policymaking process for a particular public policy and track any variations in policymakers' evidence use across levels and phases (Freeman and Maybin, 2011).

Procedure

As noted above, the specific content analysis methodology we employed was designed to track and analyze the scope, type, circumstances, and timing of evidence use by policymakers (specifically, legislators) in the formulation of federal policies to combat childhood obesity in the U.S. from 2000 to 2014. The first step in this process was the selection of relevant policy documents for analysis, including documents that reflect policy decisions (e.g., texts of public laws) and those that capture the process leading to these decisions, such as texts of bills, parliamentary hearings, and floor debates (see also Freeman and Maybin, 2011). Prior (2003) refers to this as the process by which documents are being produced and consumed in ways that are consequential to thoughts and actions in the real world. It is therefore crucial that the policy documents selected adequately represent the underlying policymaking process. In our case, policymaking closely tracks the trajectory of the legislative process in the U.S. This meant retrieving and analyzing official public documents that contain texts of bills, committee hearings and reports, floor debates, and executive actions. In addition to representing the thoughts and actions of elected officials, these documents describe the inputs provided by a range of policy stakeholders who are active in a particular policy area as well as detailed records of policy deliberations in which evidence is exchanged.

To be included in our specific study, policy documents had to be directly relevant to childhood obesity-related legislation and published between January 1, 2000 and December 31, 2014. The U.S. Government Printing Office (GPO) online database (FDsys) was selected for retrieving all relevant policy documents that met these criteria because it provides comprehensive access to official publications from all three branches of the U.S. federal government (Sproles, 2011). Given the large number of documents contained in this collection, the next logical step was to formulate and test a valid and reliable search query for retrieving all potentially relevant documents in the database that meet our inclusion criteria using a previously validated procedure (Stryker et al., 2006). In essence, this procedure involves an iterative process of formulating and implementing a search query that maximizes recall (the conditional probability that a particular document will be retrieved, given that it is relevant) and precision (the conditional probability that a particular document is relevant, given that it is retrieved). As shown in Figure 1, the search query that met these criteria was used to retrieve an initial pool of 1,888 potentially relevant documents. After two rounds of manual screening (first of title and abstract, and second of full-text), a total of 786 relevant documents were retained for analysis (224 congressional bills, 190 committee hearings and reports, and 372 records of floor debates).

Insert Figure 1 Here

The coding of each document was performed manually by a team of 14 trained undergraduate research assistants using Dedoose (2017). All coders received extensive training on the use of the codebook and the software, including examples of applying relevant codes and guided coding exercises. After recording the meta-data associated with each document (i.e., title, date, policy actor – e.g., House or Senate, and policy level – e.g., committee and/or sub-committee), their first task was to detect all incidents of evidence use defined as any reference to factual data or information as proof that a certain belief or proposition is true or valid. This definition is inclusive of research evidence and non-research evidence use, although the two are explicitly distinguished in our coding scheme. Specially, coders were instructed that ‘research evidence’ refers to empirical findings derived from systematic research methods and analyses, such as scientific experiments, observational studies (e.g., surveys, observations, and interviews), epidemiological or medical studies, research syntheses (e.g., meta-analysis, systematic reviews, and scientific expert panels), and public opinion polls that adhere to basic scientific standards (e.g., "according to the most recent data collected by CDC, nearly two out of every three Americans are overweight or obese"). In contrast, 'non-research evidence' is proof derived from personal experience; others' stories, anecdotes or testimonials; expert, practitioner or constituent opinion; and news stories (e.g., "I also notice, when I go to restaurants, the portions are larger, much less the portions that are served at home, and this is causing children to become obese"). Coders were cautioned not to confuse evidence with a claim, which is a statement or assertion that something is true without providing evidence or proof (e.g., "most American children are obese"), as claims are omnipresent in policy discourse.

In the next step, coders were tasked with coding each excerpt of text containing reference to evidence using our coding instrument. Operationally, this involved applying a binary code (yes or no) to indicate the presence or absence of each variable and each specific attribute of that variable. For example, the following excerpt is taken from a testimony given by Dr. Richard H. Carmona, the Surgeon General of the United States, in a hearing on the topic of improving child nutrition programs, held before the House Subcommittee on Education Reform of the Committee on Education and the Workforce on July 16, 2003:

> "As Surgeon General, I welcome the chance to talk with you about the public health crisis that affects every state, every city, every community, and every school across our great nation. This crisis is obesity. It's the fastest-growing cause of disease and death in America, and it's completely preventable. Nearly two out of every three Americans are overweight or obese. One out of every eight deaths in America is caused by an illness

> directly related to overweight and obesity. America's children are already seeing the initial consequences of a lack of physical activity and unhealthy eating habits. Fortunately there is still time to reverse this dangerous trend."

It is clear that Dr. Carmona is using evidence in this case to support a claim about childhood obesity being a major public health problem. Accordingly, coders were expected to apply the following codes to this excerpt: policymaking context (policy formulation); supplier (government official); type of evidence (expert testimony); source of evidence (generic, no specific source is referenced); goal of evidence use (evidence speaks to the objective status/severity of the problem); valence (neutral, evidence in not used to support or refute a particular position); motivation for using evidence (conceptual use, help audiences form an understanding of the problem).

To ensure that codes are applied correctly and reliably, coders were first asked to independently code the same three documents (a randomly selected bill, hearing, and floor debate) and their inter-coder agreement was assessed using Krippendorff's alpha (Krippendorff, 2013). Krippendorff's alpha was utilized because it can handle any number of coders; nominal, ordinal, interval, ratio, and other metrics; and in addition, missing data, and small sample sizes (Hayes and Krippendorff, 2007). Krippendorff's $\alpha \geq .80$ is typically required to demonstrate acceptable inter-coder reliability. The average alpha coefficient calculated for this initial test was .62 (SD = .88). Coders received additional training (particularly, regarding the coding of instrumental and conceptual use), and inter-coder reliability was re-assessed by having coders code a new set of three randomly selected documents; average Krippendorff's alpha was found to be acceptable after the modified training ($\alpha \geq .86$, SD = .23). In addition, we employed a quality control procedure whereby five documents coded by each coder at each phase of coding were randomly sampled and reviewed by the authors for accuracy and consistency. When systematic errors were detected, coders were asked to go back to the documents they coded and correct these errors.

Findings

We present findings we obtained from analyzing evidence use in the subset of excerpts (N = 4,684) we coded in 224 congressional bills and 190 congressional committee hearings for the purpose of demonstrating the range of questions that could be answered by applying our content analysis methodology. These two types of documents capture different phases and dynamics of the policymaking process. Bills are introduced for the purpose of drawing attention to and prioritizing public problems on policymakers' agenda. They are presented for the consideration of other policymakers, with no discussion or debate, and are a tool for garnering support around a common topic. Congressional hearings, in contrast, are focused generally on the formulation of feasible, adequate, and effective policy solutions to public problems with policymakers debating, persuading, and negotiating an agreed-upon solution and action. As a result, it is reasonable to expect that patterns of evidence use will vary between bills and committee hearings as a reflection of these differing dynamics and motivation to use evidence.

Table 1 compares use of evidence in arguments appearing in congressional bills and congressional hearings for the entire research period (2000-2014). Clearly, many more specific comparisons could be made based on the variables included in our content analysis instrument (see Appendix for a complete list), however we limit the presentation here to key comparisons regarding use of research evidence in each context. Our comparison shows that the type of research evidence used in congressional bills overwhelmingly was statistical facts (87%), e.g., "the prevalence of overweight and obesity has almost doubled among America's children and adolescents since 1980, and it is estimated that one out of five children is obese" (H.R. 2024, May 7, 2003), and that in many cases (63%) the source of statistics were not directly referenced. This likely suggests that statistical facts are perceived as an authoritative from of evidence for supporting claims. In contrast, the research evidence engaged in the context of congressional hearings was more diverse, including findings of relevant academic studies (25%) and expert testimonials (13%). In addition, a greater variety of sources of research evidence informed the work of congressional committees, most notably government research (e.g., research reports prepared by the National Center for Health Statistics, the Surgeon General, or the Centers for Disease Control and Prevention; 25% of all research evidence engaged), and to a smaller degree research produced by think tanks and foundations (6%) or by academics (5%). Importantly, anecdotal forms of evidence, such as testimonials by school cafeteria managers or personal experiences members of the committee have had with childhood obesity, are frequently presented in the context of hearings (31% of all evidence presented), suggesting that the inclusion of non-scientific evidence is integral to policymakers' evidence use in this setting.

Insert Table 1 Here

The findings summarized in Table 1 also suggest important differences in policymakers' motivations and goals of using research evidence at different phases of the policymaking process. Specifically, research evidence that is included in bills is most frequently used in conjunction with arguments that aim to establish the magnitude of the childhood obesity problem and therefore the urgency in producing a policy solution, e.g., "the epidemic growth in obesity acquired during childhood or adolescence is particularly threatening to the national health because it often persists into adulthood and increases the risk for some chronic diseases later in life" (H.R. 2024, May 7, 2003). This goal correlates with conceptual use of evidence (94% of all research evidence referenced in bills) as the intention is to influence legislators' to reach an obvious conclusion regarding the severity and urgency of the problem as a means to securing support for the proposed bill.

A different dynamics of research evidence use plays out in the context of hearings. While conceptual use of research evidence was frequent (66% of all evidence referenced), instrumental use of research evidence, or presenting evidence with the intention of supporting an argument for pursuing a specific policy or course of action, is more common than it is for bills (25% compared to 3.5%, respectively). For instance, "Over two years, SNAP [Supplemental Nutrition Assistance Program] sales have increased 335 percent within our farmers' market network, and 77 percent of Philly Food Bucks users report an increased intake of fruits and vegetables. So the evaluation has shown that Philly Food Bucks is working to encourage healthier eating and our farmers like it, too" (Senate Hearing 112-692, March 7, 2012). Tactical or political use of research is also almost exclusively observed in the context of hearings, particularly when opposing policy proposals are discussed. This is exemplified by the following excerpt taken from a House committee hearing regarding a proposed bill to improve child nutrition for disadvantaged children (H.R. 5504, July 1, 2010): "Food insecurity is a problem, but among children, it is relatively limited. For example, according to the last data we have, about 1 child in 150 will miss even a single meal in a given month because of lack of resources within the family. This and other considerable evidence suggests that all of the federal nutrition programs, food stamps, the school programs, WIC and so forth, are actually associated with increased obesity."

Insert Figure 2 Here

Next, the content analysis data we collected from analyzing policy documents over a period of 15 years can also be used to examine longitudinal patterns of policymakers' evidence use, and therefore potentially produce deeper insights in this regard. Figure 1, for example, plots the number of congressional hearings about the topic of childhood obesity in each congressional session observed in our data (107th-113th Congress) and the relative weight of instrumental use of research evidence in each session. The figure reveals a cyclical pattern of instrumental use of research evidence that generally tracks the volume of policymaking activity in each session (i.e., the number of congressional hearings held). Of note is the gradual decline in instrumental use of research evidence between the 107th and 109th Congress (2001-2006), which coincides with the first years of President George W. Bush's administration and a Republican-controlled House and Senate. The gradual decline is followed by a sharp peak through the 111th Congress (2009-2010), as control of the House and Senate shifted to Democrats and following the election into office of President Barak Obama.

Thus, as the above demonstrates, we are able to compare instrumental use of research evidence during two separate policy windows (i.e., periods of political opportunity to advance policy) where progress toward policy decisions and implementation is expected to be rapid (Kingdon, 2011). This comparison reveals that the topic of childhood obesity did not receive much attention from legislators during the first of these policy windows (as demonstrated by the handful of congressional hearings held on the topic), possibly because it was not a key issue for Republicans or because other issues, such as the 9/11 attacks on the U.S., were prioritized. In contrast, legislative interest in the topic of childhood obesity and instrumental use of research evidence in this policy context both peaked during the second policy window, not least because it was a signature topic for the Obama administration and First Lady Michelle Obama's "Let's Move" initiative (Obama, 2012). This suggests that both context and timing combine to facilitate or impede instrumental use of research evidence by policymakers. Such knowledge, in turn, can inform successful efforts to improve instrumental use of research evidence in public policymaking.

Discussion

As an object of investigation, use of evidence in public policymaking is both complex and dynamic. Policymakers routinely use a wide range of research and non-research evidence obtained from multiple actors and sources which cannot be easily or reliably tracked. In addition, there is considerable variability in how policymakers use evidence and their motivations for doing so, which vary dynamically over time (e.g., different phases of the policymaking process), contexts (e.g., different policy fields), and circumstances (e.g., political windows of opportunity). Moreover, because policymaking processes are inherently political, so is policymakers' use evidence, which is different from how they are expected to use research evidence.

In this article we argued that whereas a broad range of established methodologies are employed to study policymakers' use of evidence – surveys, interviews, observations, document analyses, social network analyses, analyses of administrative data, and experiments – they are limited in their capacity to capture and represent this complexity because they are commonly employed to track *evidence* as opposed to *use*. Moreover, when 'use' is measured, it is based on a narrow definition of this construct (i.e., how policymakers interpret and apply what they learn from research) that overlooks policymakers' strategic use of evidence for political persuasion and bargaining. Building on theories of persuasion and argumentation, we have outlined an alternative framework for tracking and analyzing use of research (and non-research) evidence in public policymaking processes which centers on arguments as the unit of analysis. Finally, we described a valid and reliable instrument we developed to study use of evidence based on the analysis of arguments in the specific policy context in which they are made, and provided an illustrative example of how it may be applied to generate a more complete and nuanced account of evidence use in public policymaking.

The data analyses we report are by no means comprehensive, but are rather intended to showcase the potential of this methodology to capture the context and timing of research evidence use, beginning with the careful selection of policy documents that closely approximate the trajectory of the policymaking process, through the recording of meta-data (i.e., time, policy actor, policy level, and nature of proceedings described in the document) and application of relevant codes, and to the multiple modes of data analysis (i.e., comparing patterns of evidence use across type of evidence, context of evidence use, policy actors and levels, and over time). In fact, our content analysis procedure can be extended to collect additional information that can support both more granular and more sophisticated analyses. For instance, by taking a record of the individuals referenced in policy documents and other potentially useful information such as their role, party or organizational affiliation, and ideological stance it is possible to compare evidence use across actors as well as to construct and study evidence networks.

We are not suggesting that the tool we developed is superior or even preferable to other available methodologies or tools for studying use of evidence. Rather, we see our content analysis method as complementing other methods to studying the use of evidence given its limitations. For one, our methodology requires significant investments of resources and time since the coding of policy documents is done manually (although given advancements in natural language processing, it may be possible in the future to automate this procedure). For another, policymakers' motivation and goal for using evidence is logically inferred from their written or spoken words rather than measured directly, and it is not always possible to corroborate these judgments, for example, by interviewing policymakers. The challenges of corroboration become a potentially serious issue if the judgments coders make are systematically biased; as we demonstrate, any potential bias can be mitigated by training coders on the task, assessing inter-coder reliability, and applying a quality control procedure. There could be other sources of bias, for example, due to the sample of documents that are available or selected for analysis. It is also obvious that not all aspects of the public policymaking process are visible in policy documents – e.g., lobbying activities, private meetings or communication, and internal discourse – and that therefore content analyzing them say nothing about use of evidence in these contexts. Recognizing the important tradeoffs involved in selecting a methodology that best matches the study's specific goals and circumstances, we are confident that the content analysis methodology we outline in this paper can produce rich and systematic account of the scope, nature, and timing of policymakers' research evidence, providing that researchers take care to select the most relevant policy documents and exercise due rigor in applying the content analysis instrument.

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Table 1: Use of Research and Non-Research Evidence in the Formulation of Federal Policies to Decrease Childhood Obesity, United States, 2000-2014

	Evidence Use in Congressional Bills (N = 1461)	Evidence Use in Congressional Hearings (N = 3223)
Type of Evidence		
Statistical facts**	87%	34%
Research study**	10%	25%
Expert opinion**	-	13%
Source of Evidence		
Generic**	63%	25%
Government Research**	22%	12%
Academic Research*	1%	5%
Think Tank Research**	-	6%
Anecdotal Evidence**	-	31%
Evidence Use Motivation		
Instrumental**	4%	25%
Conceptual**	94%	66%
Tactical**	2%	17%
Evidence Use Goal		
Objective status of problem**	74%	36%
Cause of problem**	17%	26%
Possible solution**	8%	28%
Preferred policy response**	1%	9%

Note: the statistical significance of differences in percentages between samples was estimated using a chi-square test.
* $p < .05$; ** $p < .001$.

Appendix: Content and Thematic Coding of Use of Evidence in Policy Documents

Key Variables	**Description and Codes**
Content Analysis	
Who (supplier of evidence)	Description: "supplier" of evidence is different than "source" of evidence (see below). It is the individual or entity that brings research evidence to the attention of other actors or decisionmakers. Depending on their position and role, suppliers vary in their reputation, perceived credibility and competence, political power, and capacity to transfer knowledge (O'Keefe, 2002; Davies et al., 2000). These attributes may influence the likelihood that the research evidence they share is considered in the policymaking process. Codes: legislator/policymaker; government official; researcher or scientist; practitioner or professional; lobbyist; advocate; private citizen.
What (type of evidence)	Description: "evidence" refers to research and non-research evidence as both are routinely used in the policymaking process (Baumgartner, 2013). It is important to consider that in this context, quality or accuracy of evidence may be less important than the value (emotional, political, and/or persuasive) placed on specific form of evidence (Majone, 1989). Codes: statistical fact; findings of a research study; findings of a research synthesis; findings of policy analysis; results of scientific public opinion poll; expert opinion or testimony; constituent opinion or sentiment; news story; uncorroborated evidence (e.g., from lobbyist or another policymaker); personal narrative or anecdote.
What (source of evidence)	Description: "source" refers to the producer or originator of the evidence presented. References to generic sources (e.g., "studies show") are very common in policy discourse, but in many cases the general identity of the source can be inferred (Benoit and Herzog, 2017). Credible and legitimate sources of evidence range from personal experience to studies that meet the highest standards of scientific rigor. Codes: generic; academic research; parliamentary-commissioned research; government agency research; state agency research; international agency research; foundations or non-profit (e.g., think tank) research; advocacy or lobbying entity research; industry-sponsored research; news/journalistic investigation; expert testimonial; anecdotal (personal experience-based) evidence.
Where (policymaking context)	Description: The policymaking process is the manner in which public policy is formed, implemented and evaluated and use of evidence is not limited to a particular phase. Different types of evidence may be more frequently engaged and/or carry more weight in the decisions made in each phase (Baumgartner, 2013). The phase of policymaking may be determined based on the specific content or records contained in each policy document. Codes: policy agenda-setting (e.g., bill introduction); policy formulation (e.g., legislative or investigative hearings); policy negotiation or decision

	(e.g., floor debates); policy implementation (e.g., executive actions); policy evaluation (e.g., oversight hearings).
When (timing)	Description: the date and time of the activities or proceedings described in the document (e.g., date in which a bill was introduced or committee held a meeting). This information is needed to establish the timeline of the policymaking process for a particular policy.
Thematic/Rhetorical Analysis	
Goal of evidence use	Description: evidence is most frequently used in public policymaking in combination with arguments about the objective status of a problem, its causes or consequences, preferred policy response, and accountability. The content of the argument suggests how evidence is interpreted and/or framed (Scheufele and Tewksbury, 2007). Codes: evidence speaks to the objective status of the problem (e.g., magnitude, burden); evidence speaks to the cause(s) of the problem; evidence speaks to parties who should be held accountable for causing the problem (e.g., personal vs. social responsibility); evidence speaks to probable or preferred policy solutions to the problem; evidence speaks to who should be held accountable for addressing the problem; evidence speaks to the evaluation of past or current policy response to the problem (e.g., as part of oversight).
Valence of evidence use	Description: A primary function of evidence use in policy discourse is to support or refute arguments about the problem or policy considered (Majone, 1989). Some arguments are two-sided (i.e., include both pros and cons) and tend to be more persuasive when counterarguments are noted and then refuted (O'Keefe, 2002). Codes: pro, con, two-sided, or neutral.
Motivation for evidence use	Description: Use of evidence in policymaking processes involves different motivations (Weiss, 1979). *Instrumental use* occurs when evidence is used to directly inform decisions about policies or actions. *Conceptual use* refers to situations in which evidence is presented to influence how policymakers understand issues, problems, or potential solutions. *Tactical use*, also called political and symbolic use, occurs when evidence is used to justify predetermined positions or policy solutions. It also involves use of evidence to accelerate or slow down policy decisions, for example, by increasing or decreasing ambiguity regarding the problem or solution. The content of the arguments attached to evidence can be used to infer the supplier's underlying motivation for using evidence in a particular instance. Codes: instrumental use; conceptual use (scope and nature of problem); conceptual use (causes of problem); conceptual use (feasible policy solutions for problem); tactical use (political, justifying or undermining an existing stance or proposal); tactical use (political, establishing accountability for solving the problem); tactical use (strategic, increasing or decreasing ambiguity about the problem/solution); tactical use (symbolic, making connections to core social values or norms such as national security or social justice).

Figure 1: Summary of Document Selection Procedure

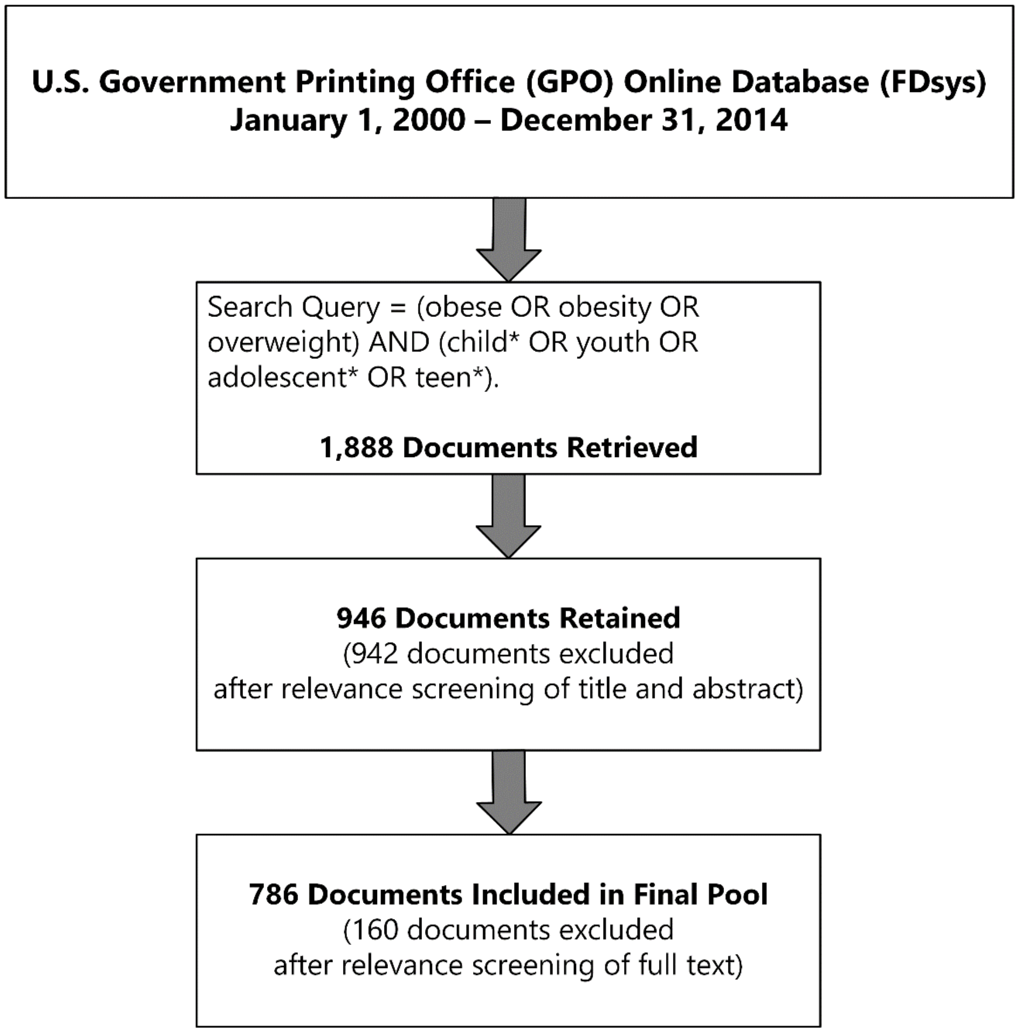

Figure 2: Instrumental Use of Research Evidence in Congressional Hearings Regarding Federal Policies to Decrease Childhood Obesity, United States, 2000-2014